

FORD GOBIKE SYSTEM

DATA EXPLORATION

Exploratory Data Analysis

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PROJECT OVERVIEW

OBJECTIVES

- 01** Analyze customer riding behavior
- 02** Identify trip trends
- 03** Understand user demographics
- 04** Discover business insights
- 05** Recommend strategies for improving bike sharing services

DATASET OVERVIEW

Include

- Total Records
- Total Columns
- Dataset Source
- Time Period

The dataset contains trip-level information including trip duration, stations, user demographics and membership details.



PEAK HOUR ANALYSIS

Key Insights

- Most bike trips occur during morning and evening commuting hours.
- Demand is significantly higher during office hours compared to late-night hours.
- The usage pattern indicates that the bike-sharing service is primarily used for daily commuting.

Chart - 5

```
# Chart - 5 visualization code
plt.figure(figsize=(12,6))

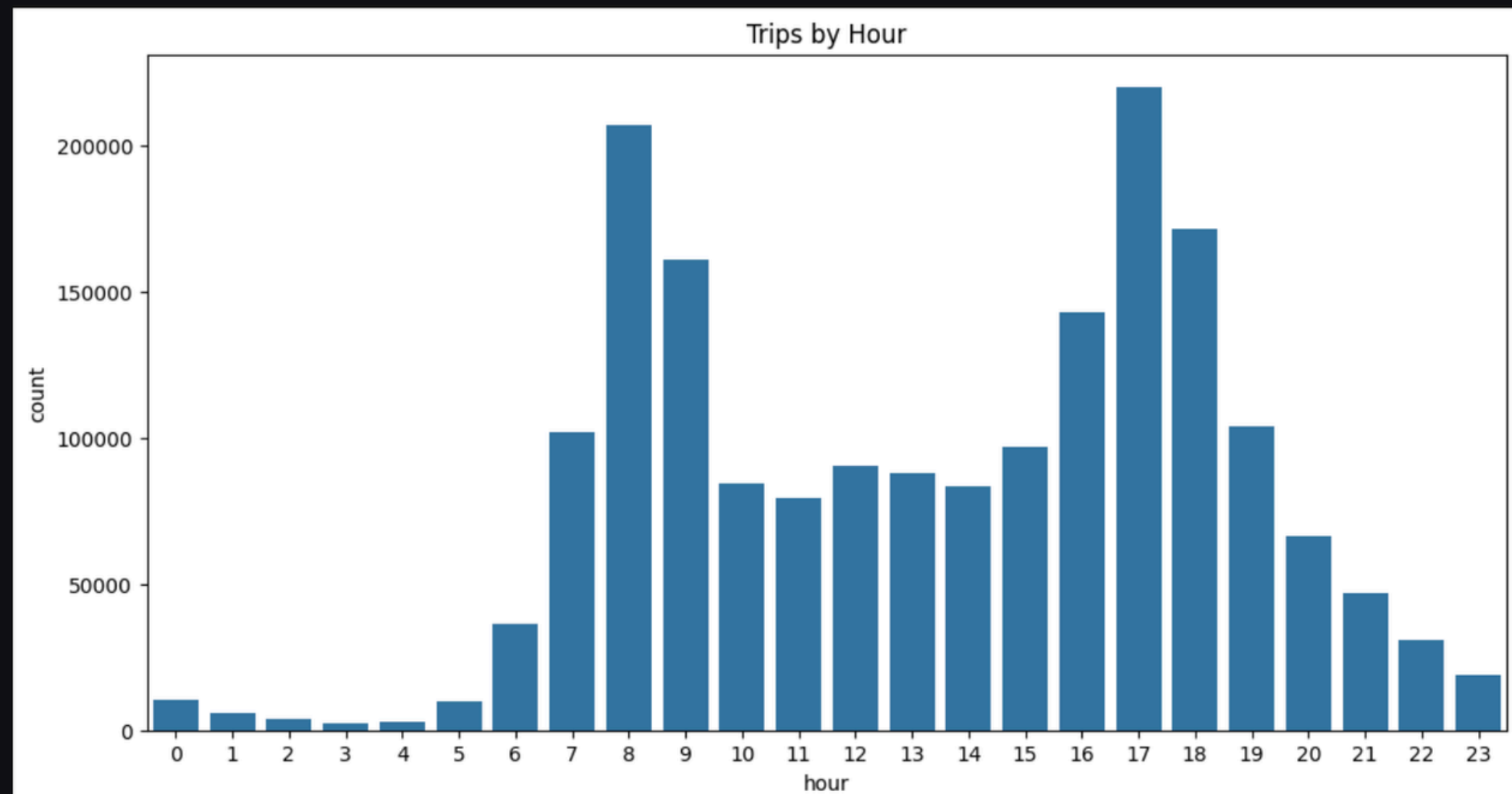
sns.countplot(data=bike_df,x='hour')

plt.title('Trips by Hour')

plt.show()
```

[257] ✓ 2.8s

Python



USER TYPE DISTRIBUTION

Key Insights

- Subscribers account for the majority of total bike trips.
- Customers represent a much smaller share of users.
- Regular membership plays a major role in the success of the bike-sharing service.

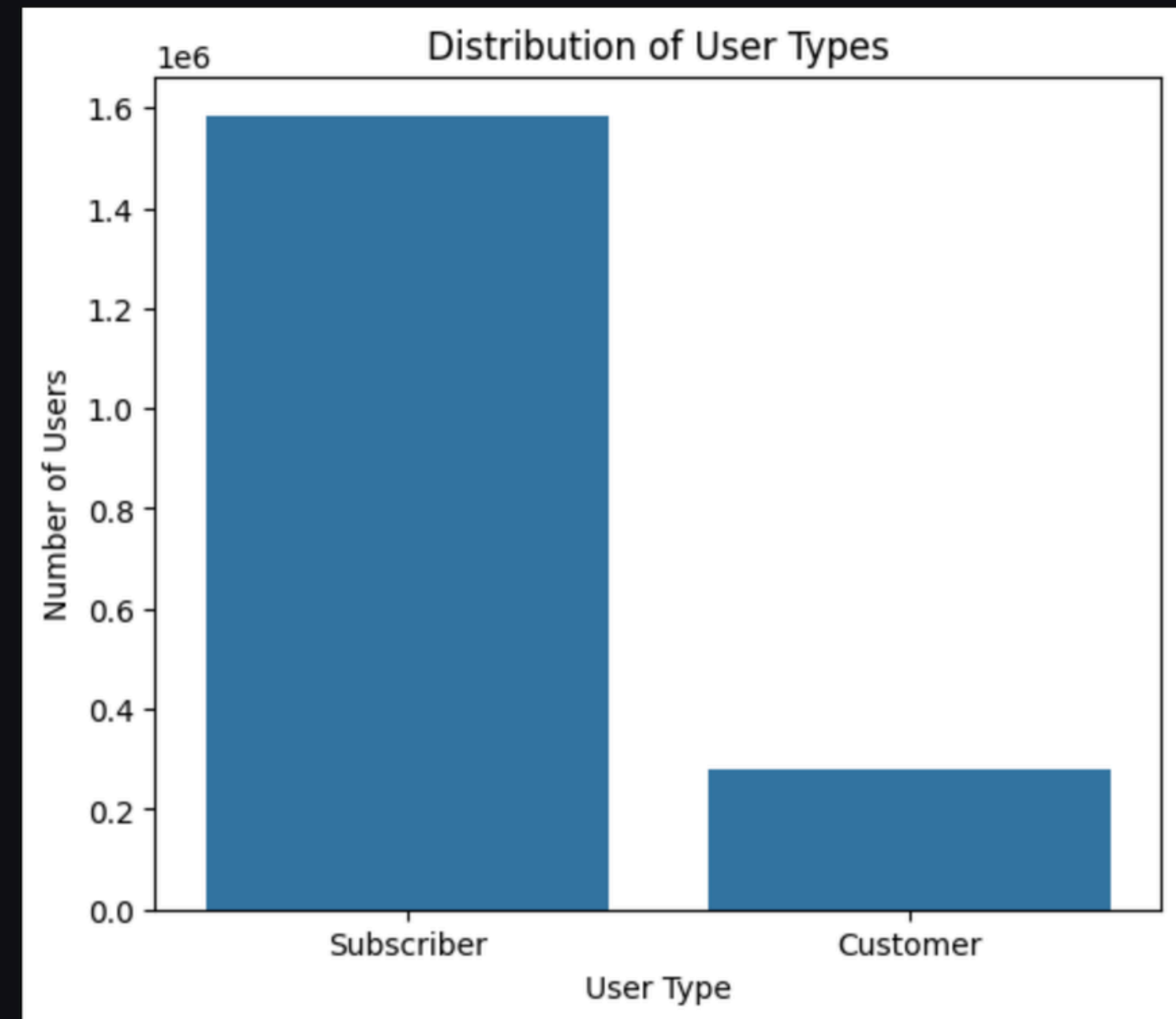
```
# Chart - 2 visualization code
plt.figure(figsize=(6,5))

sns.countplot(data=bike_df, x='user_type')

plt.title('Distribution of User Types')
plt.xlabel('User Type')
plt.ylabel('Number of Users')

plt.show()

✓ 2.8s
```



MONTHLY TRIP TREND

Key Insights

- Bike usage varies across different months of the year.
- Seasonal changes influence the demand for bike-sharing services.
- Monthly demand trends can help optimize bike allocation and maintenance schedules.

Chart - 4



```
# Chart - 4 visualization code
plt.figure(figsize=(12,6))

sns.countplot(data=bike_df,
              x='month',
              order=['January', 'February', 'March', 'April', 'May', 'June',
                    'July', 'August', 'September', 'October', 'November', 'December'])

plt.xticks(rotation=45)

plt.title('Trips by Month')

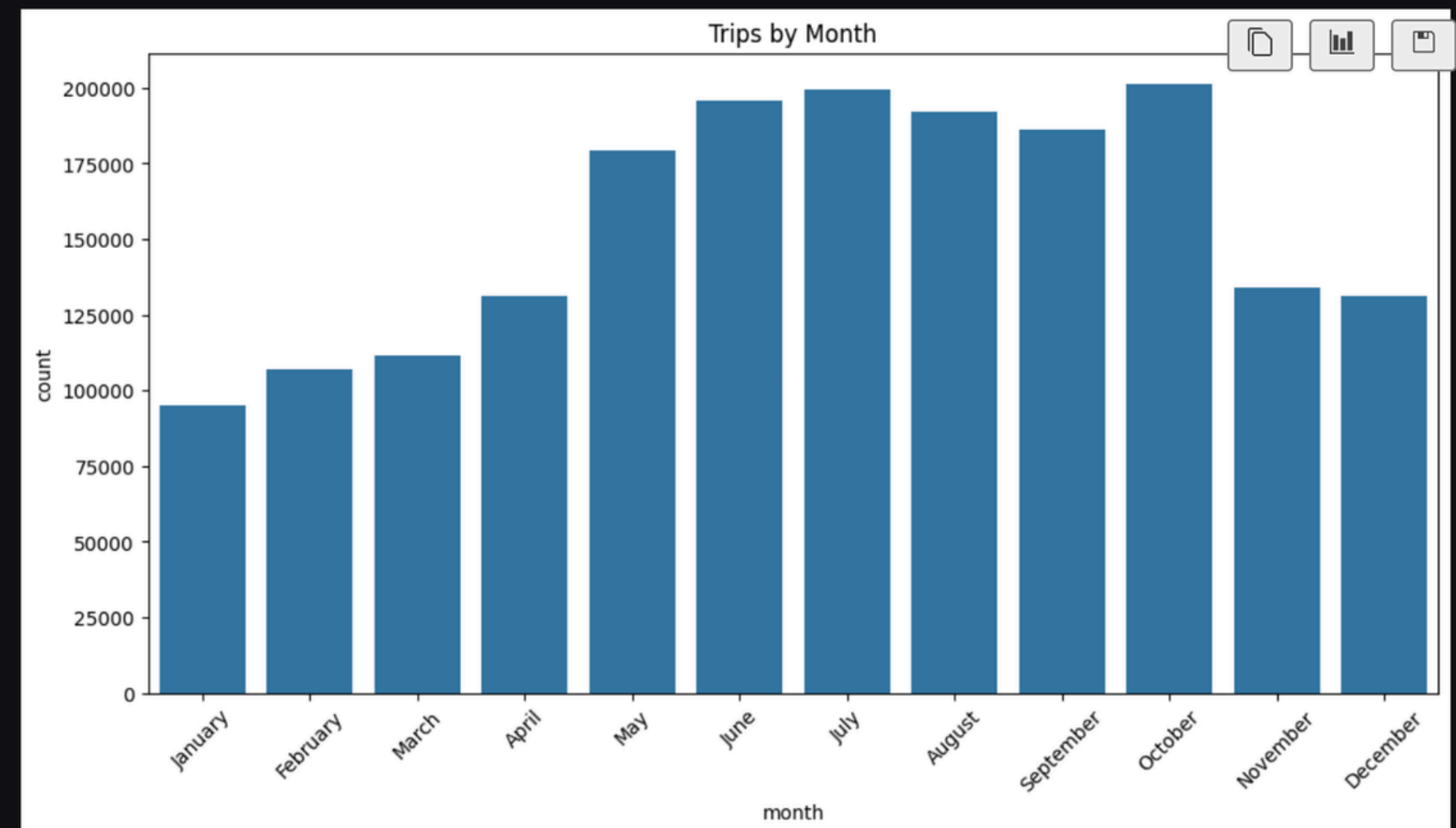
plt.show()
```

[256]

✓ 2.3s

Python

...



MOST POPULAR START STATIONS

Key Insights

- A few stations generate a large proportion of total trips.
- Popular stations experience consistently high customer demand.
- These stations require higher bike availability and frequent maintenance.

Chart - 15 - Pair Plot

```
# Pair Plot visualization code
top_station = bike_df['start_station_name'].value_counts().head(10)

plt.figure(figsize=(12,6))

sns.barplot(x=top_station.values,
            y=top_station.index)

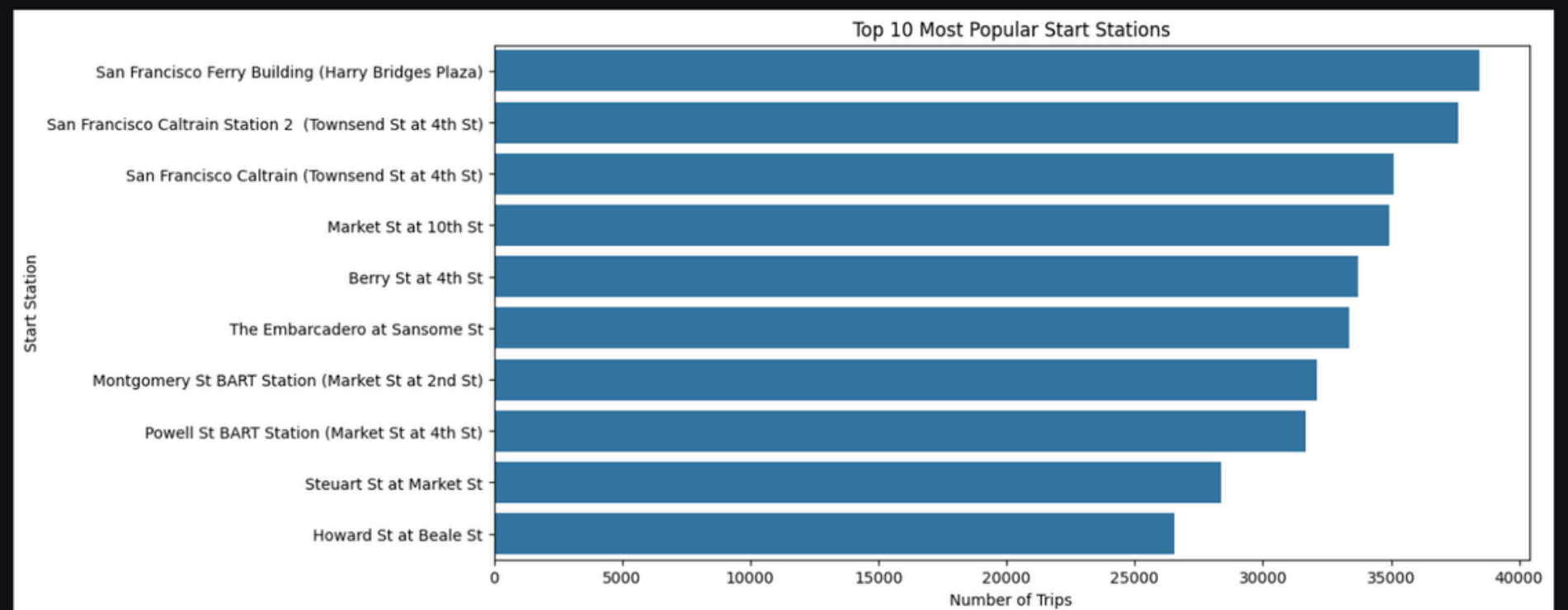
plt.title('Top 10 Most Popular Start Stations')
plt.xlabel('Number of Trips')
plt.ylabel('Start Station')

plt.show()
```

[268]

✓ 0.1s

Python



RIDER AGE DISTRIBUTION

Key Insights

- Most riders belong to the young adult age group.
- The service is primarily used by working professionals and students.
- Younger riders form the largest customer segment of the bike-sharing system.

Chart - 8

```
# Chart - 8 visualization code
bike_df['age'] = 2018 - bike_df['member_birth_year']

plt.figure(figsize=(10,6))

sns.histplot(bike_df['age'], bins=30)

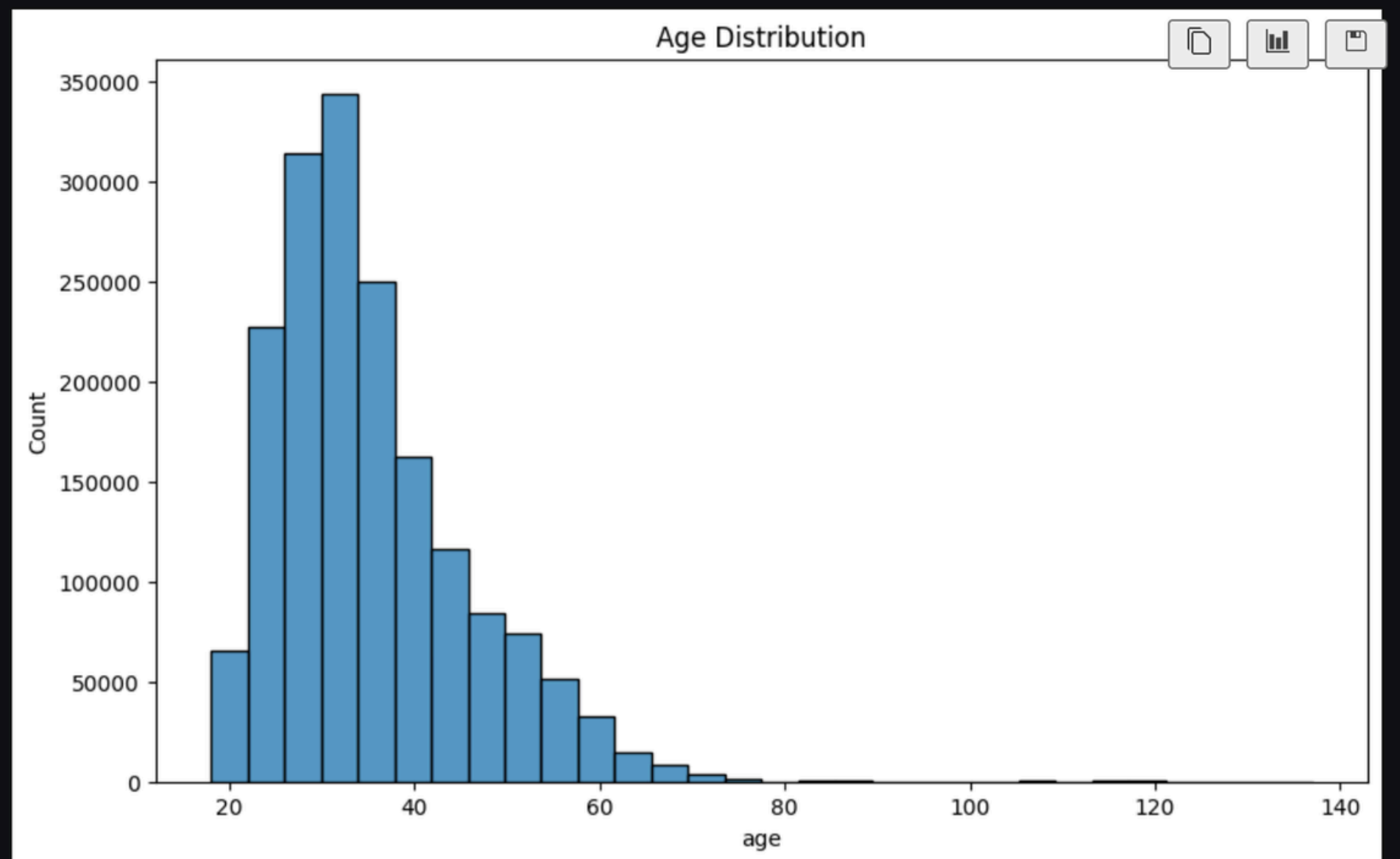
plt.title("Age Distribution")

plt.show()
```

[261]

✓ 1.3s

Python



BUSINESS INSIGHTS

Key Business Findings

- **Subscribers generate the majority of total bike trips.**
- **Most rides are short, indicating the service is mainly used for daily commuting.**
- **Peak demand occurs during morning and evening office hours.**
- **A small number of stations contribute to the highest trip volume.**
- **Young adults are the primary users of the Ford GoBike system.**
- **Monthly demand varies, indicating seasonal usage patterns.**
- **Data-driven resource allocation can significantly improve operational efficiency.**

BUSINESS RECOMMENDATIONS

- **Increase bike availability during peak commuting hours.**
- **Expand bike capacity at high-demand stations.**
- **Introduce promotional offers to convert casual customers into subscribers.**
- **Promote the Bike Share for All program to increase participation.**
- **Schedule maintenance during low-demand periods.**
- **Use monthly demand trends to improve bike distribution.**
- **Continuously analyze customer behavior to support data-driven decision-making.**

THANK YOU